Reconstructed Atkin interface: all 21 pass

2026-10-27 | T-5518 | ATKIN-NATIVE-GETALL-01 and -02

Observations only. All 666 native-body observations repeat byte for byte. In each run, all 299 rational parameters succeed at both arithmetic widths: **598 successful cohort GETALL calls, no cohort zero divisions, all 21 curves represented.** The historical selector remains unidentified.

Original routine bodies run with newly written arithmetic, root-enumeration and kernel-validation adapters and modern coefficient tables. The original generation caller remains unrecovered.

The fixed cohort contains 11 reconstructed omissions and 10 publications. Models are y^2 = x^3 - 3x + b over Fp. Exact fields, coefficients, orders, UIDs, seeds and ordinals are bound in rows.json; no visitation or historical rejection event is inferred.

RNG stream	Bits	Omitted	Published	Parameters	Checks
863711343	239	5	3	123	189
863966734	192	2	3	70	98
871002724	521	0	1	14	22
871696354	256	1	1	28	43
872952608	384	0	1	10	16
891607927	224	3	1	54	65

Measured counts per width and per run. All parameters succeed. The input contains 315 curve/level cells at 15 type-II levels, 53 through 293. A cell without a rational parameter is not a failed curve.

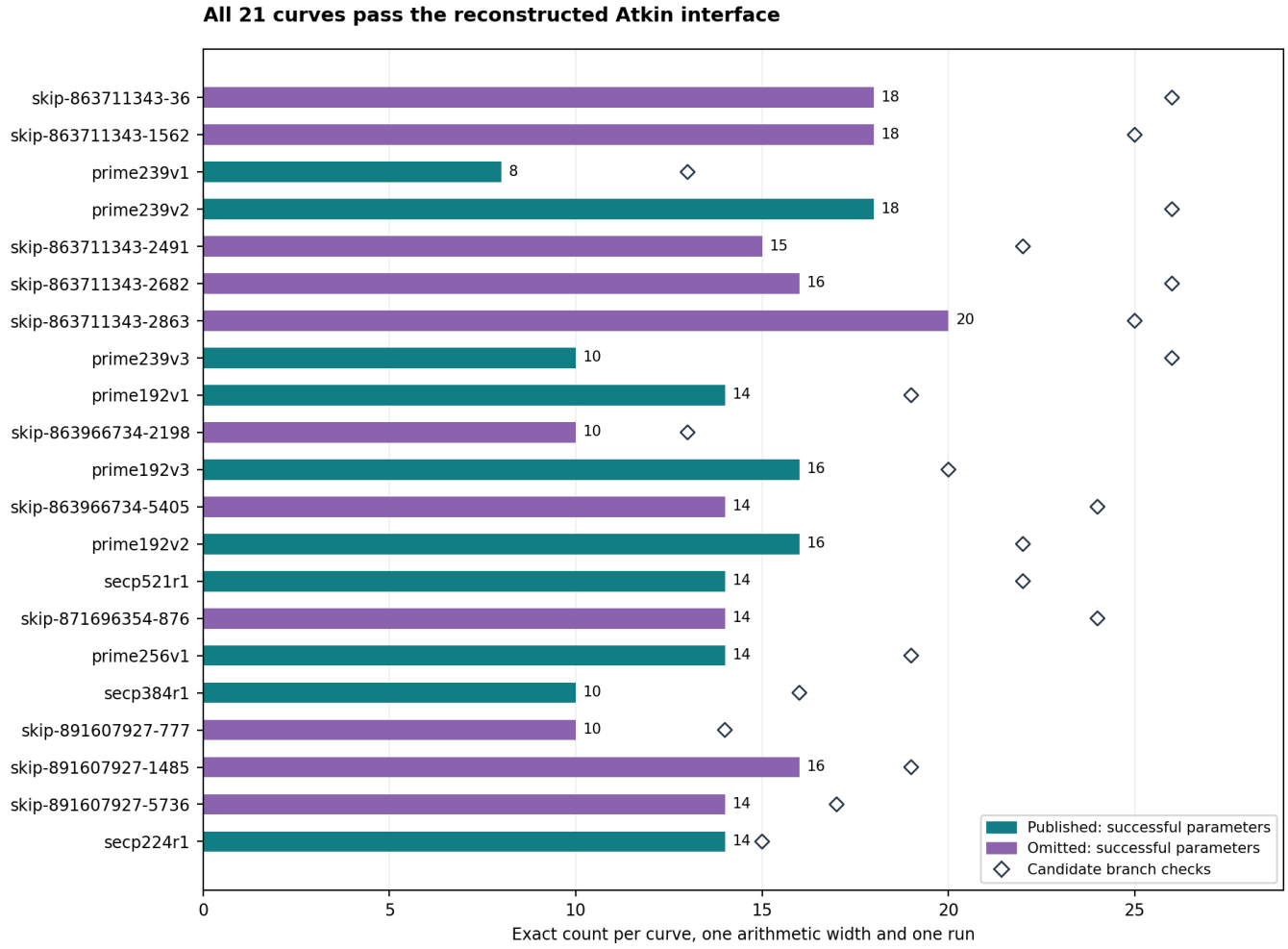
What the recovered source does. The author-hosted note dated March 5, 1993 includes actual GETPAA / GETALL bodies, compiled here without changes. It constructs neighboring models from derivative ratios:

```
E4 = -A/3; E6 = -B/2; r = E6/E4
a_prime = r * (j Psi_j) / Psi_alpha
rq = a_prime * Psi_alpha(alpha,jq) / (ell jq Psi_j(alpha,jq))
E4q = rq^2 * jq / (jq-1728); E6q = rq * E4q
Aq = -3 ell^4 E4q; Bq = -2 ell^6 E6q
```

This interface does not extract a cube root of j. On every cohort model, A = -3 gives E4 = 1. That observation concerns this recovered interface; other historical field-dependent paths remain unestablished.

Independent arithmetic check. PARI/GP matched all 433 cohort candidate branches and all 299 accepted kernels to the recorded Frobenius traces. The native routine and reconstructed validator received no order or trace. The complete repeat passes its formal export check; its relocated archive independently replays.

Exact curve-level observations



ATKIN-NATIVE-GETALL-01 / -02 | 299 parameters, 433 branch checks per width | exact finite census; no sampling interval

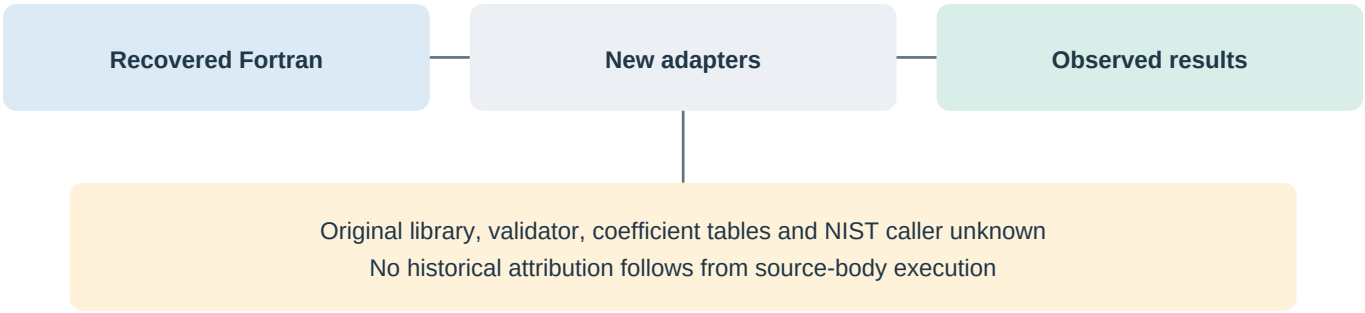
All 21 curves, 299 parameters and 433 branch checks per width and per run. These are exact finite counts; no population interval or historical timing estimate is inferred. Fields, b values and UIDs are in curve-summary.csv and the sealed rows.json.

A count-only collision. Omitted 863711343/36 and published P-239v2 share the same field and the same aggregate tuple: 18 parameters, 26 branch checks, zero divisions. Any rule depending only on that tuple assigns the same outcome to both.

Omitted P-224 ordinal 777 uses 10 parameters and 14 checks; the publication uses 14 and 15. A budget monotone solely in these two totals cannot reject the omitted candidate while retaining the publication. Per-level profiles, real historical runtimes and arbitrary combined rules are outside this statement.

Controls per run: 60 derivative calls (15 tables, two fields, two widths, five outputs each); four regular GETALL calls; two known exceptional GETALL calls; and two after-cohort fixture repeats. The ordinary model is (p,A,B,j,ell) = (1009,990,660,34,53). The exceptional model (1009,199,469,24,53), alpha=98, triggers the expected zero division. Both widths also pass multiword representation checks.

Historical source and dependency boundaries



The original ZL library, BLVNOC, GETJQN and CHARLA were not recovered. Newly written adapters provide exact GMP arithmetic using normalized MSW-first digits, complete sorted FLINT neighbor enumeration, and corrected CCR/Newton kernel checks. The source ZL representation is not established. Modern coefficient tables are not authenticated period tables. ELKCOF, eigenvalue/CRT stages and the original curve-generation caller are not executed.

Source-history correction. The February 5, 1997 LiDIA 1.3 announcement lists prime-field point counting as work in progress. The currently author-linked UVW package is from 2024. The ja93 preprint is the February 21, 1995 version of the same Morain article. None supplies the missing 1997 implementation.

Failures retained. A pre-run link failure was repaired with an SDK library path. A GP comment syntax error kept the first audit at FAIL despite its completion marker; a comment-only repair passes. The first formal export had 686 encoding/identity findings. Its records are unchanged. A prospectively frozen, complete repeat with 333 embedded-input cases per width passes with zero findings. Both runs have 666 native-body calls and identical observations.

Evidence custody. The two exact archives, all failures, independent replay scripts and source hashes are retained. Original author source and the native binary are private because redistribution permission is unestablished. Both canonical store uploads returned HeadObject403; Git archives use the existing PR exception. Merge readiness is not asserted. Diagnostic native process times were 53.8456 and 54.5844 seconds on a shared desktop, with no timing comparison.

Next source gap: recover an original validator, coefficient-table set or actual generation caller; test any resulting mechanism against the unchanged 11 omissions and 10 publications. Existing CCR and modular-coordinate diagrams were checked and remain unchanged; this round adds no official scientific decision.

Atkin, March 5, 1993 (retrieved; exact hashes retained)

Atkin, March 3, 1993 (retrieved; exact hashes retained)

LiDIA 1.3 announcement (retrieved; exact hashes retained)

Morain author preprint (retrieved; exact hashes retained)

Evidence: BUNDLE-FIRST.json; BUNDLE-REPEAT.json; CROSS-REPEAT.json; EXPORT-CHECK.json; INDEPENDENT-AUDIT-02.json; archive replay receipts. Cohort SHA256: 93ac4774b993a994a17d9327df92e0f2b287e66755cd933fdc683015fec1d545.